Dr. Anima Kujur

Postdoctoral Researcher
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Research Profile

Postdoctoral researcher working at the intersection of scientific machine learning, dynamical systems, and biomedical data science, with applications in neuroscience and medical imaging. My work focuses on developing reliable, interpretable, and scalable models for high-dimensional scientific data (MRI, fMRI, EEG/LFP). I combine deep learning with operator-theoretic approaches (e.g., Koopman methods) to better understand model behavior and underlying system dynamics. Beyond predictive performance, I study robustness, generalization, uncertainty, and failure modes, with the goal of building models that are both accurate, interpretable, and scientifically meaningful.

Research Experience

Postdoctoral Researcher

Interdisciplinary Center for Scientific Computing (IWR), Heidelberg University, Germany
Oct 2024 - Present
Supervisor: Dr. Zahra Monfared

- Develop novel machine learning models for nonlinear dynamical systems combining deep learning with operator-theoretic methods
- Design long-horizon prediction reconstruction models for neural signals (EEG/LFP)
- Investigate model reliability, stability, and failure modes in sequence models
- Analyze latent representations and interpretability using Koopman-based frameworks
- Contribute to method-driven research with applications in computational neuroscience

Senior Research Fellow (SRF)

Jawaharlal Nehru University, New Delhi, India
Jan 2022 – Mar 2024

- Developed deep learning models for computer vision tasks in medical imaging (MRI, CT)
- Designed and evaluated segmentation and classification frameworks for clinical diagnosis
- Conducted systematic studies on data complexity, robustness, and model performance
- Improved reliability and generalization of imaging models under varying data conditions

Visiting Research Scholar

University Medical Center Mannheim, Heidelberg University, Germany
Oct 2023 – Dec 2023

- Developed machine learning models for multimodal MRI analysis
- Worked on radiogenomics (MGMT methylation prediction in glioblastoma)
- Integrated multi-source imaging features for improved predictive modeling

Junior Research Fellow (JRF)

Jawaharlal Nehru University, New Delhi, India
Jul 2019 – Dec 2021

- Built strong foundations in machine learning, optimization, and numerical methods
- Conducted research in deep learning for medical image analysis

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Research Internship

IWR, Heidelberg University, Germany
Aug 2019 – Sep 2019

- Modeled physical systems using ODEs and numerical approximation techniques
- Developed early experience in scientific computing and simulation-based modeling

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Education

Ph.D., Computer Science & Technology
Jawaharlal Nehru University, New Delhi, India (2019–2024)

Master of Computer Application (MCA)
Jawaharlal Nehru University, New Delhi India (2016–2019)

Master of Science (Applied Mathematics)
University of Hyderabad, Hyderabad, India (2013–2016)

Publications (Selected)

- Learning High-Dimensional Transient Neural Dynamics for Zero-Shot Cross-Subject Reconstruction, ICML Workshop (2026)
- Multimodal Deep Learning for Dynamic and Static Neuroimaging: Integrating MRI and fMRI for Alzheimer's Disease Analysis, Journal of Machine Learning for Modeling and Computing, (in production) 2026
- Transient Neural Dynamics Reconstruction, NeurIPS Workshop (2025)
- Autoregressive ConvLSTM for fMRI Forecasting, NeurIPS Workshop (2025)
- High-Fidelity Radiological Image Segmentation with Deep Learning, IEEE AIC (2025)
- DUNET: Dilated U-Net for Brain Tumor Segmentation, Defence Life Science Journal (2024)
- Data Complexity Based Evaluation of Brain MRI Classification, IEEE Access (2022)

Scientific Presentations

Invited Talk

- ICBRA 2025, Prague
  Multimodal Neuroimaging and Koopman-Based Neural Dynamics Analysis

Conferences

- AICOMAS 2025 (Paris)
- Multiple international ML & AI conferences

Posters

- DKFZ Postdoc Symposium 2026
- ICML 2026 workshop
- COSYNE 2026
- NeurIPS 2025 Workshops

Supervision & Teaching

- PhD Mentor (One currently)
- Supervised MSc-level research projects in:
  - fMRI time-series modeling
  - Interpretable AI
  - Clinical machine learning applications
- Tutor at international summer school (AIMS, Cape Town)
- Maths Teacher (High School: 11th & 12th Grade in India)

Workshop & Conference Participations

- AI for Science workshop, Helmholtz Munich Campus, (Munich, 2026)
- Grand Challenge Conference, Berlin University Alliance, (Berlin, 2026)
- AI in Medicine (Sorbonne-Heidelberg, 2025)
- Integrative Think Tank (Heidelberg, 2025)
- Infectious Disease Modeling Workshop (SWIM, Heidelberg, 2025)
- HeiAIMS Summer School (Cape Town, South Africa, 2024)

Core Competencies

- **Machine Learning & Scientific AI**
  Supervised and unsupervised learning, representation learning, data-driven modeling

- **Computer Vision & Medical Imaging**
  MRI/fMRI analysis, image segmentation, multimodal learning

- **Reliable & Interpretable AI**
  Model robustness, uncertainty estimation, explainability, failure mode analysis

- **Dynamical Systems & Time-Series Modeling**
  Koopman operator theory, DMD/EDMD, nonlinear dynamics, long-horizon prediction

- **Scalable Machine Learning**
  Efficient model design, large-scale training workflows

- **Programming & Scientific Computing**
  Python (PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn)

Soft Skills

- Structured and goal-oriented working style
- Strong analytical and problem-solving ability
- Interdisciplinary thinking and adaptability
- Effective communication and collaboration skills
- Independent and self-driven, with high sense of responsibility
- Ability to quickly learn and work in new domains
- Mentoring and inspiring

Languages

- English: Fluent (spoken and written)
- German: Basic (A2 level)
- Hindi: Native

Interests & Activities

- Travelling and exploring new cities and cultures
- Interest in spirituality and personal growth
- Hiking, Cycling and outdoor activities
- Basketball and team sports
- Dance and creative expression
- Cooking

References

Dr. Michael Winckler Michael.Winckler@iwr.uni-heidelberg.de (Advisor)

Dr. Zahra Monfared zahra.monfared@iwr.uni-heidelberg.de (Postdoc PI)

Dr. Shervin Safavi shervin.safavi@cmclab.org (Research Collaborator)

Prof. Zahid Raza zahidraza@mail.jnu.ac.in (PhD Supervisor)

Prof. Jürgen Hesser Juergen.Hesser@medma.uni-heidelberg.de (Internship Supervisor)